Randomized and low-rank approximation

Sensitivity-dependent row sampling for ℓ_p embeddings when p exceeds two

Difficulty: hard **Importance:** interesting to the community

Status: source-stated conjecture; no resolution found in screening on 2026-09-08.

Fix $p > 2$. For a full-column-rank matrix $A \in \mathbb{R}^{n\times d}$, with rows a_i^T, set

$$s_i = \sup_{x\neq 0} \frac{|a_i^T x|^p}{\|Ax\|_p^p}, \qquad \mathcal{S} = \sum_{i=1}^n s_i.$$

For a scalar $\alpha > 0$, retain row i independently with probability

$$q_i = \min\{1, 1/n + s_i/\alpha\},$$

and scale each retained row by $q_i^{-1/p}$; let $\widetilde{A}$ be the resulting matrix.

Do constants $C_p, c_p > 0$ exist such that, for every A and $0 < \varepsilon, \delta < 1/2$, one can choose α with

$$\sum_i q_i \le C_p \varepsilon^{-2}(\mathcal{S} + d)\log^{c_p}\left(\frac{2nd}{\varepsilon\delta}\right),$$

while, with probability at least $1-\delta$,

$$(1-\varepsilon)\|Ax\|_p^p \le \|\widetilde{A}x\|_p^p \le (1+\varepsilon)\|Ax\|_p^p \qquad \text{for all } x \in \mathbb{R}^d?$$

The size is the expected number of retained rows. The question concerns this independent sampling rule based on the ordinary ℓ_p sensitivities, not arbitrary row weights or augmented sensitivities.

This would exploit easy instances through total sensitivity rather than using the worst-case dimension-only bound. It applies directly to sketching ℓ_p regression and other computations on the column space of A. The p>2 ordinary-sensitivity theorem of Woodruff–Yasuda has size $\widetilde{O}_p(\varepsilon^{-2}\mathcal{S}^{2-2/p})$.

References

1. D. P. Woodruff and T. Yasuda, *Sharper Bounds for ℓ_p Sensitivity Sampling*. ICML 2023, arXiv:2306.00732v2. Definition 1.1 fixes the sampling model, Theorem 1.5 gives the p>2 bound, and Section 3 explicitly conjectures $\widetilde{O}(\varepsilon^{-2}(\mathcal{S}+d))$. Paper.
2. A. Munteanu and S. Omlor, *Optimal bounds for ℓ_p sensitivity sampling via ℓ_2 augmentation*. ICML 2024, PMLR 235, pp. 36769–36796. Question 1.2 and Section 1.1 distinguish the solved/obstructed p≤2 situation from the remaining p>2 case. Paper.
3. R. Chhaya, A. Dasgupta, D. Feldman, and S. Shit, *Deterministic Coreset for Lp Subspace*. arXiv:2601.00361 (2026). This claimed deterministic embedding result was withdrawn on May 15, 2026 because some proofs are incomplete; it is cited only to prevent a stale abstract from being mistaken for a resolution. Paper.

Status check — 2026-09-08

Searched “p>2 sensitivity sampling conjecture 2025 2026”, “sensitivity sampling S+d”, and the exact paper titles; checked the current arXiv records. No resolution of the stated p>2 rule was found. Results using ℓ_2 augmentation for p≤2 do not settle it. The 2026 deterministic embedding claim was withdrawn (latest v3, May 15, 2026), so its abstract is not evidence that any open case has been resolved.